

Polarization of Gravitational Waves From Turbulence: Tension Between Simulation and Analytic Approaches

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I. GRAVITATIONAL WAVES FROM A TURBULENT SOURCE

Before specializing to particular spectral and temporal models, we collect the standard analytic derivation of the gravitational-wave (GW) spectrum sourced by turbulence [1–5]. This fixes the conventions used throughout and, in particular, shows why the remaining sections reduce the problem to computing the contracted source correlator $H_{ijij}(\mathbf{k}, \omega)$ of Eq. (10) on the diagonal $\omega = k$.

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A. Sourced wave equation

Write the tensor perturbations of a spatially flat FLRW background as

$$ds^2 = a^2(\tau)[-d\tau^2 + (\delta_{ij} + h_{ij})dx^i dx^j], \quad \partial_i h_{ij} = h_{ii} = 0, \quad (1)$$

with h_{ij} transverse-traceless (TT). The linearized Einstein equations give

$$h''_{ij} + 2\frac{a'}{a}h'_{ij} - \nabla^2 h_{ij} = 16\pi G a^2 \Pi_{ij}^{\text{TT}}, \quad (2)$$

where $' = d/d\tau$ and Π_{ij}^{TT} is the TT part of the anisotropic stress. For an incompressible, non-relativistic turbulent fluid the source is the Reynolds stress,

$$\Pi_{ij}^{\text{TT}}(\mathbf{k}, \tau) = \Lambda_{ij,lm}(\hat{k}) T_{lm}(\mathbf{k}, \tau), \quad T_{lm} = w v_l v_m, \quad \Lambda_{ij,lm} = P_{il}P_{jm} - \frac{1}{2}P_{ij}P_{lm}, \quad (3)$$

with $w = \bar{\rho} + \bar{p}$ the enthalpy density, $P_{ij} = \delta_{ij} - \hat{k}_i \hat{k}_j$, and Λ the TT projector. The double TT projection together with the Gaussian (Wick) reduction of the quartic velocity correlator collapses the two-point function of (3) onto the contracted form $R_{ijij} = R_{ii}R_{jj} + \frac{1}{3}R_{ij}R_{ij}$ that appears in Eq. (10).

Turbulence acts on eddy-turnover times short compared with a Hubble time, so over the source lifetime we drop the expansion ($a \approx 1$, friction negligible) and Eq. (2) becomes, in Fourier space, a forced oscillator,

$$h''_{ij}(\mathbf{k}, \tau) + k^2 h_{ij}(\mathbf{k}, \tau) = 16\pi G \Pi_{ij}^{\text{TT}}(\mathbf{k}, \tau). \quad (4)$$

B. Green-function solution and the GW spectrum

The retarded solution of Eq. (4) and its time derivative are

$$h_{ij}(\mathbf{k}, \tau) = \frac{16\pi G}{k} \int_{\tau_i}^{\tau} d\tau' \sin[k(\tau - \tau')] \Pi_{ij}^{\text{TT}}(\mathbf{k}, \tau'), \quad \dot{h}_{ij}(\mathbf{k}, \tau) = 16\pi G \int_{\tau_i}^{\tau} d\tau' \cos[k(\tau - \tau')] \Pi_{ij}^{\text{TT}}(\mathbf{k}, \tau'). \quad (5)$$

With the Isaacson energy density $\rho_{\text{GW}} = \langle \dot{h}_{ij} \dot{h}_{ij} \rangle / (32\pi G)$ and the unequal-time correlator (UETC) of the TT stress,

$$\langle \Pi_{ij}^{\text{TT}}(\mathbf{k}, \tau_1) \Pi_{ij}^{\text{TT}*}(\mathbf{k}', \tau_2) \rangle = (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{k}') \Pi(k; \tau_1, \tau_2), \quad (6)$$

inserting (5) and averaging the product $\cos[k(\tau - \tau_1)] \cos[k(\tau - \tau_2)] \rightarrow \frac{1}{2} \cos[k(\tau_1 - \tau_2)]$ over the fast GW oscillations gives the master formula

$$\frac{d\rho_{\text{GW}}}{d \ln k} \simeq \frac{2G k^3}{\pi} \int d\tau_1 \int d\tau_2 \cos[k(\tau_1 - \tau_2)] \Pi(k; \tau_1, \tau_2). \quad (7)$$

The GW spectrum is thus the temporal Fourier transform of the source UETC, evaluated at the on-shell frequency $\omega = k$. For a statistically stationary (or quasi-stationary) source, $\Pi(k; \tau_1, \tau_2) = \Pi(k; \tau_1 - \tau_2)$ and the double time integral factorizes into the source duration T_{src} times the temporal transform $\tilde{\Pi}(k, \omega) = \int d(\Delta\tau) e^{i\omega\Delta\tau} \Pi(k; \Delta\tau)$, which is precisely the spatio-temporal correlator $H_{ijij}(\mathbf{k}, \omega)$ of Eq. (10). Hence

$$\frac{d\rho_{\text{GW}}}{d \ln k} \propto k^3 H_{ijij}(\mathbf{k}, \omega)|_{\omega=k} \implies \Omega_{\text{GW}}(p) \propto p^3 H(p, p), \quad p = \frac{k}{k_0}, \quad q = \frac{\omega}{k_0}, \quad (8)$$

the GW dispersion $\omega = k$ becoming the diagonal $q = p$ used in Sec. V (with characteristic strain $h_c^2(p) = p H(p, p)$). Everything downstream is therefore an evaluation of H_{ijij} , whose spatial part is a convolution of the velocity power spectrum (quadratic source, Eqs. (11)–(12)) and whose temporal part is the decorrelation model carried by $\Pi(k; \tau_1, \tau_2)$.

C. Spectral shape from the literature

For Kolmogorov inertial-range input the analytic results have three robust pieces:

- *Infrared* ($k \ll k_0$): causality forces the stress correlator to be analytic (white) as $k \rightarrow 0$, so $H \rightarrow \text{const}$ and $\Omega_{\text{GW}} \propto k^3$ [3, 4].
- *Peak* ($k \sim k_0$): set by the largest eddies, with parametric amplitude $(d\Omega_{\text{GW}}/d\ln k)_{\text{peak}} \sim (H_*/k_0) \Omega_K^2$, i.e. the squared kinetic-energy fraction suppressed by the source-duration factor $H_*/k_0 \sim H_*\tau_{\text{eddy}}$ [1, 3].
- *Ultraviolet* ($k \gg k_0$): *model-dependent*, fixed by the temporal decorrelation $\Pi(k; \tau_1, \tau_2)$. Kraichnan sweeping gives the Gogoberidze tail [2], while a δ -correlated or a decaying-turbulence kernel give different slopes.

The UV slope is precisely where the four UETC models derived below (δ -spectrum vs. white-noise, Kraichnan vs. decay) differ, which is the focus of the remainder of the paper.

II. KRAICHNAN-K41 MODEL (GOGOBERIDZE 2007)

We use the spatial/temporal Fourier pair $R_{ij}(\mathbf{r}, \tau) = \int \frac{d^3k}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{r}} F_{ij}(\mathbf{k}, \tau)$ and $f(\eta_k, \tau) = \int \frac{d\omega}{2\pi} e^{-i\omega\tau} \tilde{f}(\eta_k, \omega)$, and adopt the isotropic incompressible model of [2] (Eqs. 23–26),

$$F_{ij}(\mathbf{k}, \tau) = A(k) P_{ij}(\hat{k}) f(\eta_k, \tau), \quad P_{ij}(\hat{k}) = \delta_{ij} - \hat{k}_i \hat{k}_j, \quad (9)$$

with $A(k) = C_k \epsilon^{2/3} k^{-11/3} / 4\pi$, $f(\eta_k, \tau) = \exp(-\frac{\pi}{4} \eta_k^2 \tau^2)$, and $\eta_k = \frac{1}{\sqrt{2\pi}} \epsilon^{1/3} k^{2/3}$.

The fourth-rank Fourier-space correlator $H_{ijkl}(\mathbf{k}, \omega) = (2\pi)^{-4} \int d\tau d^3\xi e^{i\omega\tau - i\mathbf{k}\cdot\xi} R_{ijkl}(\xi, \tau)$ with the contraction $R_{ijij} = R_{ii}R_{jj} + \frac{1}{3}R_{ij}R_{ij}$ (Gogoberidze Eq. 28) reads, in real space,

$$H_{ijij}(\mathbf{k}, \omega) = \frac{1}{(2\pi)^4} \int d\tau d^3\mathbf{r} e^{i\omega\tau - i\mathbf{k}\cdot\mathbf{r}} \left[R_{ii}(\mathbf{r}, \tau) R_{jj}(\mathbf{r}, \tau) + \frac{1}{3} R_{ij}(\mathbf{r}, \tau) R_{ij}(\mathbf{r}, \tau) \right]. \quad (10)$$

Inserting the spatial Fourier representation of R_{ij} , the $\mathbf{r}$ integral produces a momentum-conserving $\delta^{(3)}(\mathbf{k} - \mathbf{q} - \mathbf{p})$. Renaming $\mathbf{q} \rightarrow \mathbf{k}_1$, $\mathbf{u} = \mathbf{k} - \mathbf{k}_1$,

$$H_{ijij}(\mathbf{k}, \omega) = \frac{1}{(2\pi)^7} \int d\tau e^{i\omega\tau} d^3k_1 \left[F_{ii}(\mathbf{k}_1, \tau) F_{jj}(\mathbf{u}, \tau) + \frac{1}{3} F_{ij}(\mathbf{k}_1, \tau) F_{ij}(\mathbf{u}, \tau) \right]. \quad (11)$$

The temporal Fourier transform then converts the τ integral into a frequency δ -function, leaving the frequency convolution

$$H_{ijij}(\mathbf{k}, \omega) = \frac{1}{(2\pi)^8} \int d^3k_1 d\omega_1 \left[\tilde{F}_{ii}(\mathbf{k}_1, \omega_1) \tilde{F}_{jj}(\mathbf{u}, \omega - \omega_1) + \frac{1}{3} \tilde{F}_{ij}(\mathbf{k}_1, \omega_1) \tilde{F}_{ij}(\mathbf{u}, \omega - \omega_1) \right]. \quad (12)$$

Substituting Eq. (9) and using $P_{ii}(\hat{k}) = 2$ together with $P_{ij}(\hat{k}_1) P_{ij}(\hat{u}) = 1 + (\hat{k}_1 \cdot \hat{u})^2$, the tensor contractions reduce Eq. (12) to

$$H_{ijij}(\mathbf{k}, \omega) = \frac{1}{(2\pi)^8} \int d^3k_1 d\omega_1 A(k_1) A(u) \tilde{f}(\eta_{k_1}, \omega_1) \tilde{f}(\eta_u, \omega - \omega_1) \left[4 + \frac{1}{3} (1 + (\hat{k}_1 \cdot \hat{u})^2) \right]. \quad (13)$$

This is the Fourier representation valid for all k ; we treat $\mathbf{k} > \mathbf{k}_0$ and $\mathbf{k} \leq \mathbf{k}_0$ separately below.

A. $\mathbf{k} > \mathbf{k}_0$

The law of cosines $\hat{k}_1 \cdot \hat{u} = (k^2 - k_1^2 - u^2)/(2k_1u)$ turns the bracket in Eq. (13) into a homogeneous polynomial in (k, k_1, u) ; expanding and grouping gives the geometric kernel

$$\mathcal{K}(k, k_1, u) \equiv \frac{27}{6} + \frac{k^4}{12k_1^2u^2} + \frac{k_1^2}{12u^2} + \frac{u^2}{12k_1^2} - \frac{k^2}{6u^2} - \frac{k^2}{6k_1^2} \quad (\text{cf. Eq. 29 of [2]}). \quad (14)$$

The temporal transform $\tilde{f}(\eta_k, \omega) = (2/\eta_k) \exp(-\omega^2/\pi\eta_k^2)$ combines with $A(k)$ into the single spectral factor

$$g(k, \omega) \equiv A(k) \tilde{f}(\eta_k, \omega) = \frac{2E_k}{4\pi k^2 \eta_k} \exp\left(-\frac{\omega^2}{\pi\eta_k^2}\right), \quad E_k = C_k \epsilon^{2/3} k^{-5/3}. \quad (15)$$

Going to spherical coordinates for $\mathbf{k}_1$ ($d^3k_1 = k_1^2 dk_1 d\phi d\mu$ with $\mu = \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}_1$), the azimuthal integral contributes 2π . Trading μ for $u = \sqrt{k^2 + k_1^2 - 2kk_1\mu}$ via $d\mu = u du / (kk_1)$, with $u \in [|k - k_1|, k + k_1]$,

$$H_{ijij}(\mathbf{k}, \omega) = \frac{1}{(2\pi)^7} \int_0^\infty \frac{k_1 dk_1}{k} \int_{|k-k_1|}^{k+k_1} u du \int_{-\infty}^\infty d\omega_1 g(k_1, \omega_1) g(u, \omega - \omega_1) \mathcal{K}(k, k_1, u). \quad (16)$$

Switching to the dimensionless variables $q = k_1/k$, $s = u/k$, $\Omega = \omega/(\epsilon^{1/3}k^{2/3})$, $\Omega_1 = \omega_1/(\epsilon^{1/3}k^{2/3})$, g becomes $g(k, \omega) = (1/\sqrt{2\pi}) C_k \epsilon^{1/3} k^{-13/3} \exp(-2\Omega^2)$. The product $g(k_1, \omega_1)g(u, \omega - \omega_1)$ thus separates into a dimensional prefactor $\propto k^{-26/3}$ times a manifestly dimensionless Gaussian envelope $\Phi(q, \Omega_1; s, \Omega - \Omega_1) = q^{-13/3} s^{-13/3} \exp[-2\Omega_1^2 q^{-4/3} - 2(\Omega - \Omega_1)^2 s^{-4/3}]$. Collecting all k -dependent factors gives the scaling form

$$H_{ijij}(\mathbf{k}, \omega) = \frac{C_k^2 \epsilon}{256 \pi^8} k^{-5} \mathfrak{H}(\Omega), \quad \mathfrak{H}(\Omega) = \int_0^\infty dq q \int_{-\infty}^\infty d\Omega_1 \int_{|1-q|}^{1+q} ds s \Phi \tilde{\mathcal{K}}(q, s), \quad (17)$$

with $\tilde{\mathcal{K}}(q, s) = \mathcal{K}(k, k_1, u)|_{k_1=qk, u=sk}$ (homogeneous of degree zero). All physical scales are absorbed in the prefactor $C_k^2 \epsilon k^{-5}$.

B. Gogoberidze (2007) Appendix A form

Using the Gaussian-erfc identity (Eq. A.1 of [2], $\int_0^\infty dy e^{-Ay^2 - B(y-x)^2} = \sqrt{\pi/[4(A+B)]} e^{-ABx^2/(A+B)} \text{erfc}[-Bx/\sqrt{A+B}]$), the ω_1 integration in Eq. (16) can be done in closed form, yielding the doubly-bounded form

$$H_{ijij}(k, \omega) = \frac{\varepsilon}{24\pi(2\pi)^{3/2}k} \int_{k_0}^{k_d} dk_1 k_1^{-10/3} \int_{\max(|k_1-k|, k_0)}^{\min(k_1+k, k_d)} du u^{-10/3} (k_1^{-4/3} + u^{-4/3})^{-1/2} \times \left[27 - \frac{k^2}{k_1^2} - \frac{k^2}{u^2} + \frac{k^4}{2k_1^2 u^2} + \frac{k_1^2}{2u^2} + \frac{u^2}{2k_1^2} \right] \exp\left(-\frac{2\varepsilon^{-2/3}\omega^2}{k_1^{4/3} + u^{4/3}}\right) \text{erfc}\left(-\frac{\sqrt{2}\varepsilon^{-1/3}\omega}{(k_1^{-4/3} + u^{-4/3})^{1/2}}\right). \quad (18)$$

k_0 is the outer scale and k_d the dissipation scale; the integration window collapses near the boundaries of the inertial range. This integral is finite as $k \rightarrow 0$.

Define the dimensionless parameters

$$M = \left(\frac{\varepsilon}{k_0}\right)^{1/3}, \quad \frac{k_d}{k_0} = R^{3/4}, \quad p = \frac{k}{k_0}, \quad q = \frac{\omega}{k_0}, \quad x = \left(\frac{k_1}{k_0}\right)^{-4/3}, \quad y = \left(\frac{u}{k_0}\right)^{-4/3}, \quad (19)$$

where $M = (\varepsilon/k_0)^{1/3}$ is the outer-scale turbulent Mach number (sound speed set to unity), equal to the rms eddy velocity u_0 . The Mach number enters the GW kernel through the k -dependent sweeping rate $\eta_k = \frac{1}{\sqrt{2\pi}} \varepsilon^{1/3} k^{2/3} = \frac{M}{\sqrt{2\pi}} k_0^{1/3} k^{2/3}$, which couples frequency to wavenumber in the Gaussian factor of Eq. (18). As a result M sets the *effective* wavenumber support of the integral: the exponential $\exp[-2q^2/(M^2(z^{4/3} + y^{4/3}))]$ (with $z = k_1/k_0$) cuts off contributions with $z^{4/3} + y^{4/3} \lesssim 2q^2/M^2$, so the peak of the GW spectrum shifts with M . The substitution $k_1 = k_0 x^{-3/4}$, $u = k_0 y^{-3/4}$ gives $dk_1 k_1^{-10/3} = -(3/4)k_0^{-7/3} x^{3/4} dx$ and $(k_1^{-4/3} + u^{-4/3})^{-1/2} = k_0^{2/3} (x + y)^{-1/2}$; the integration limits become $x \in [R^{-1}, 1]$ and $y \in [y_{\min}(x), y_{\max}(x)]$ with

$$y_{\max}(x) = \min(|x^{-3/4} - p|^{-4/3}, 1), \quad y_{\min}(x) = \max((x^{-3/4} + p)^{-4/3}, R^{-1}). \quad (20)$$

Plugging into Eq. (18) yields

$$H_{ijij}(p, q) = \frac{3M^3 k_0^{-4}}{256(2\pi)^{3/2} p} \int_{R^{-1}}^1 dx x^{3/4} \int_{y_{\min}}^{y_{\max}} dy y^{3/4} (x + y)^{-1/2} \times \left[27 - p^2(x^{3/2} + y^{3/2}) + \frac{p^4}{2} x^{3/2} y^{3/2} + \frac{1}{2}(x^{-3/2} y^{3/2} + y^{-3/2} x^{3/2}) \right] \times \exp\left(-\frac{2xy}{x+y} \frac{q^2}{M^2}\right) \text{erfc}\left(-\frac{\sqrt{2}q}{M\sqrt{x+y}}\right). \quad (21)$$

C. Mach-dependent peak of the stationary spectrum

Evaluating the kernel (21) on the GW dispersion $\omega = k$ (the diagonal $q = p$ of Eq. (8)) gives the dimensionless energy-density spectrum $\Omega_{\text{GW}}(p) \propto p^3 H_{ijij}(p, p; M, R)$ of stationary Kraichnan-K41 turbulence. The Mach number

enters *only* through the sweeping rate $\eta_k \propto M k^{2/3}$, i.e. through the Gaussian $\exp[-\frac{2xy}{x+y} q^2/M^2]$ and the accompanying erfc in Eq. (21). This factor is a soft cutoff that suppresses the integrand once $p \gtrsim M$, so the spectral peak tracks the Mach number, $p_{\text{peak}} \sim M$, until it saturates near the outer scale at large M . Figure 1 shows this directly: the peak slides from $p_{\text{peak}} \simeq 0.05$ at $M = 0.03$ to $p_{\text{peak}} \simeq 2.6$ at $M = 3$, while the amplitude climbs steeply (the M^3 prefactor of Eq. (21) times the widening M -support of the integral). The infrared rise is the universal causal slope $\Omega_{\text{GW}} \propto p^3$ (Sec. IC), common to all M ; only the location and height of the peak respond to M .

The M -dependence of the IR amplitude is fixed entirely by the prefactor: for $p \ll M$ the Gaussian and erfc factors of Eq. (21) reduce to unity, so $H_{ijij}(p \rightarrow 0)$ is set by the M^3 prefactor alone and $\Omega_{\text{GW}} \rightarrow C M^3 p^3$ with C a universal (M -independent) constant. Equivalently the characteristic strain scales as $h_c = \sqrt{pH} \sim M^{3/2} \sqrt{p}$ in the IR. Compensating the spectrum by M^{-3} (the strain by $M^{-3/2}$) therefore collapses every curve onto the single universal p^3 line in the IR, isolating the genuinely M -dependent peak and UV; Fig. 1(b) confirms this collapse numerically (residual spread $< 1.5\%$ at $p \simeq 3 \times 10^{-4}$, and a measured IR exponent $d \ln H / d \ln M \rightarrow 3.00$ as $p \rightarrow 0$).

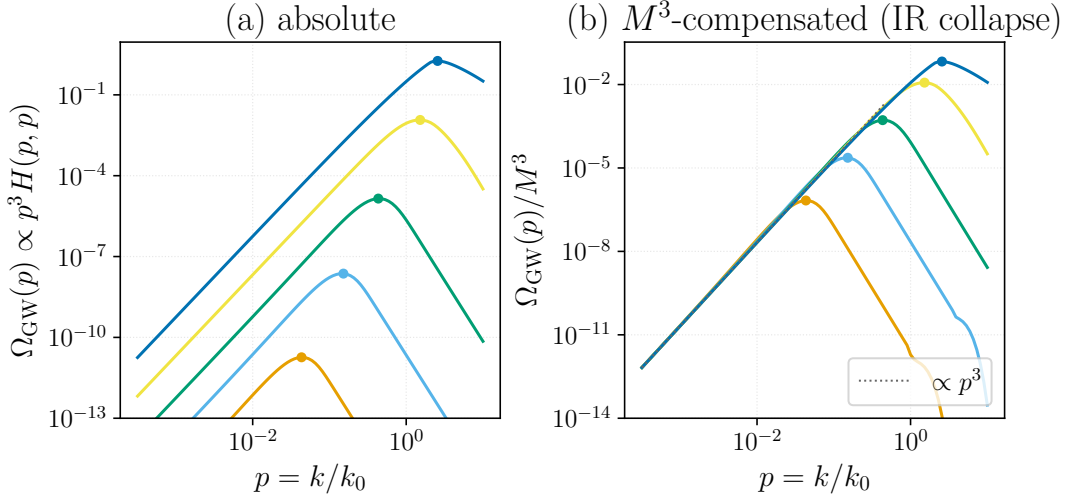


FIG. 1. Stationary Kraichnan-K41 (Gogoberidze 2007) GW spectrum $\Omega_{\text{GW}}(p) \propto p^3 H_{ijij}(p, p; M, R)$, computed from Eq. (21) on the on-shell diagonal of Eq. (8), at $R = 10^4$. The five curves are $M = 0.03, 0.1, 0.3, 1, 3$, ordered by increasing amplitude and peak frequency (lowest, leftmost curve is $M = 0.03$; highest, rightmost is $M = 3$); dots mark each peak. (a) Absolute: shared IR slope p^3 , amplitude growing $\sim M^3$, peak moving right with M ($p_{\text{peak}} \sim M$ in the inertial range, bending over near the outer scale). (b) M^3 -compensated, Ω_{GW}/M^3 : since the IR amplitude scales as M^3 (strain as $M^{3/2}$), all curves collapse onto the universal p^3 line (dotted) in the IR and peel off only at their M -dependent peak. Generated by `Notebooks/stationary_mach_peak.py`.

D. Spectra in the $\Omega(k)/k$ convention: comparison with simulations

To compare the analytic kernel with the direct numerical simulations of Roper Pol et al. [6] we recast the spectra in their convention. There $\Omega(k)$ denotes the energy *per logarithmic wavenumber*, $\Omega(k) = d\rho/d \ln k$, so that $\Omega(k)/k = E(k)$ is the ordinary per- dk spectrum. Our master result $\Omega_{\text{GW}}(p) \propto p^3 H_{ijij}(p, p)$ of Eq. (8) is precisely $d\rho_{\text{GW}}/d \ln k$, hence

$$\frac{\Omega_{\text{GW}}(k)}{k} \propto \frac{\Omega_{\text{GW}}(p)}{p} = p^2 H_{ijij}(p, p; M, R), \quad p = \frac{k}{k_0}, \quad (22)$$

with H_{ijij} the stationary kernel of Eq. (21). Our derivation of Ω_{GW} assumes a *Kolmogorov inertial range* only; we therefore plot the analytic GW spectrum solely for $k \geq k_0$ (above the spectral peak) and add no Batchelor infrared tail or other component that the derivation does not justify.

Figure 2 shows this analytic spectrum together with the slope $k^{-11/3}$ that the simulation of [6] (run ini2) reports for its GW inertial range, drawn as a reference guide; we plot no simulation data points here and defer a quantitative data-level comparison to future work. Above the peak the analytic spectrum steepens *continuously*: its local slope is shallower than $-11/3$ just above the peak and passes through $-11/3$ (equivalently $\Omega_{\text{GW}} \propto k^{-8/3}$, the Gogoberidze 2007 value) only near $k \sim 10\text{--}40 k_0$, before the k -dependent sweeping decorrelation—a cutoff at $k \sim M^3 k_0$ —bends it below any single power law. The analytic stationary kernel therefore does *not* reproduce the *extended* $k^{-11/3}$ range

seen in the simulation; it merely crosses that slope. The crossing is itself transonic-specific (for smaller M the cutoff sits closer to the peak, for larger M the near-peak slope is shallower), so the kernel possesses no single, M -independent inertial slope. This high- k disagreement is precisely the simulation–analytic tension that motivates this paper.

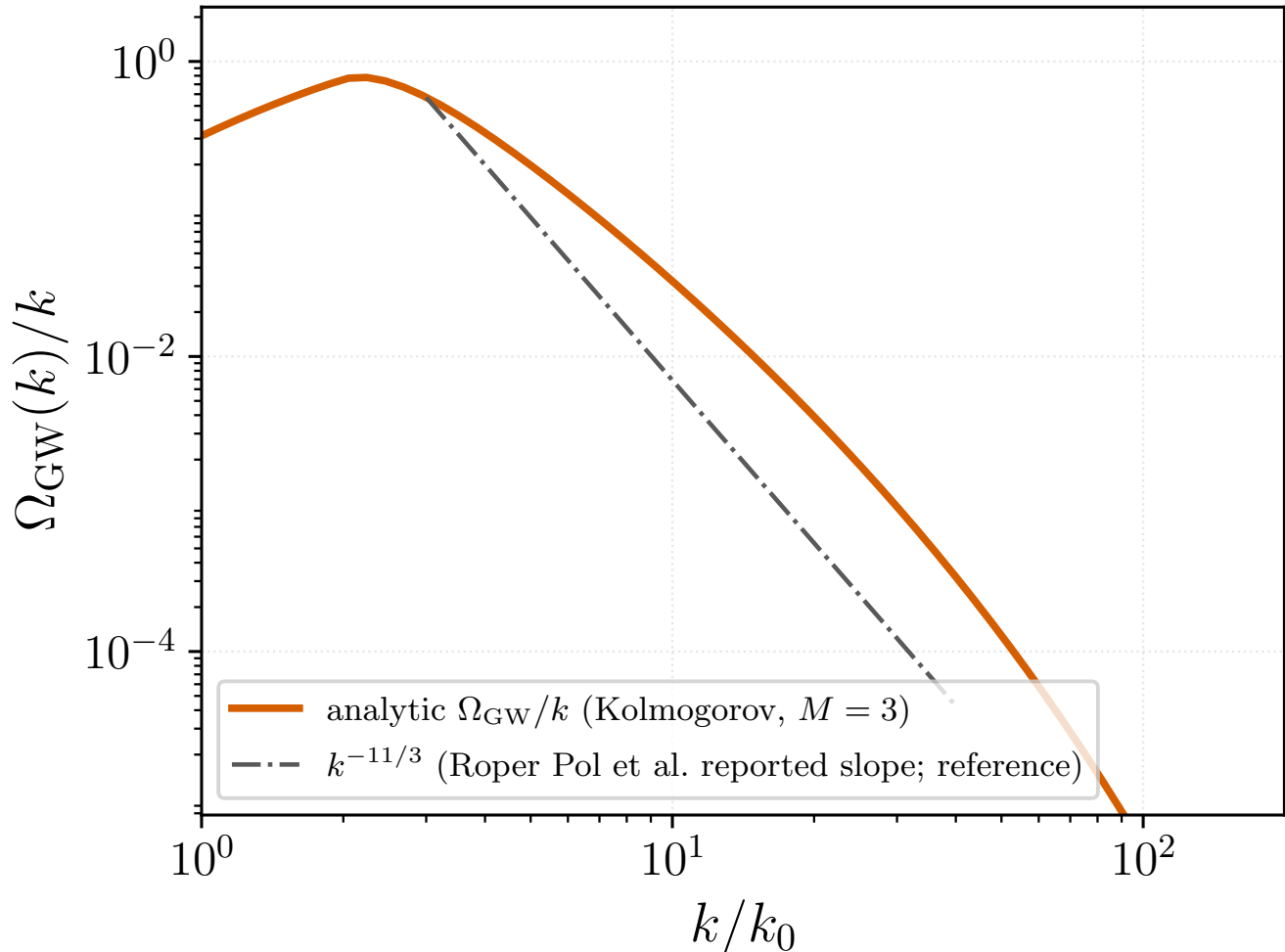


FIG. 2. Analytic GW spectrum in the Roper Pol et al. [6] convention $\Omega(k)/k = E(k)$ (energy per logarithmic wavenumber divided by k). Solid line: analytic GW spectrum $\Omega_{\text{GW}}(k)/k = p^2 H_{ijij}(p, p; M, R)$ [Eq. (22), stationary kernel Eq. (21), $M = 3$, $R = 10^4$], shown only over the Kolmogorov inertial range $k \geq k_0$. Dash-dot line: the $k^{-11/3}$ slope reported by the simulation (run ini2), as a reference only—no simulation data points are plotted. The analytic spectrum steepens continuously through the sweeping cutoff ($k \sim M^3 k_0$) and crosses, rather than follows, the $k^{-11/3}$ reference. Generated by `Notebooks/roperpol.convention.spectra.py`.

E. Discussion: the Mach number, equivalent definitions, and the spectral shape

The preceding two subsections make the Mach number the single control parameter of the stationary spectrum, so it is worth stating precisely what M is, why the conflicting definitions in the literature are the *same* quantity, and what about the spectrum does and does not respond to it.

a. Explicit formulation. Throughout we use

$$M = \left(\frac{\varepsilon}{k_0} \right)^{1/3} = u_0, \quad (c_s \equiv 1), \quad (23)$$

the outer-scale eddy velocity in units of the sound speed (Eq. (19)). The grouping is fixed by dimensions: with $[\varepsilon] = L^2 T^{-3}$ and $[k_0] = L^{-1}$ one has $\varepsilon/k_0 = (L/T)^3$, so $(\varepsilon/k_0)^{1/3}$ is a velocity and the inverse grouping $(k_0/\varepsilon)^{1/3}$ —

which appears occasionally in the literature — is an *inverse* velocity and cannot be a Mach number. M does not enter Eq. (18) as a free amplitude; it enters *only* through the k -dependent sweeping rate $\eta_k = \frac{1}{\sqrt{2\pi}} \varepsilon^{1/3} k^{2/3} = \frac{M}{\sqrt{2\pi}} k_0^{1/3} k^{2/3}$, hence only through the Gaussian $\exp[-\frac{2xy}{x+y} q^2/M^2]$ and the accompanying erfc of Eq. (21).

b. Two definitions, one quantity. The literature quotes the Mach number either as a velocity ratio $M = u_{\text{rms}}/c_s$ or as the cascade combination $(\varepsilon/k_0)^{1/3}$ [2]. For fully developed Kolmogorov turbulence these are the same quantity: setting $c_s = 1$ and using the inertial-range relation $\varepsilon \sim u_0^3 k_0$ turns the first into the second. The equivalence rests on three assumptions: (i) a stationary, constant-flux cascade with a well-defined ε across an inertial range of width $k_d/k_0 = R^{3/4} \gg 1$; (ii) energy concentrated at the outer scale, which the $E(k) \propto k^{-5/3}$ slope guarantees because $\int_{k_0}^{\infty} E dk = \frac{3}{2} C_K (\varepsilon/k_0)^{2/3}$ converges at k_0 ; and (iii) absorbing the resulting $\mathcal{O}(1)$ constant, so that $u_{\text{rms}} = \sqrt{\frac{3}{2} C_K} (\varepsilon/k_0)^{1/3} \approx (1.5\text{--}2) (\varepsilon/k_0)^{1/3}$ depending on C_K and on whether one normalises $\int E dk$ to $\langle u^2 \rangle$ or $\frac{1}{2} \langle u^2 \rangle$. The bare $(\varepsilon/k_0)^{1/3}$ is thus the single-eddy outer velocity, smaller than the integrated rms by this convention-dependent factor — the dominant source of numerical disagreement between papers for the same physical flow. The two definitions *diverge* when the assumptions fail: for non-Kolmogorov spectra in which the energy integral is ultraviolet-dominated (the white-noise / Batchelor model of Sec. III B, $E \propto k^2$), where $(\varepsilon/k_0)^{1/3}$ no longer measures u_{rms} ; for decaying turbulence, where $\varepsilon \rightarrow \varepsilon(t)$ and $M \rightarrow M(t)$ (Sec. IV); and in MHD, where the relevant ratio is often the Alfvénic $M_A = u_{\text{rms}}/v_A$ rather than the acoustic M [3, 7].

c. Shape versus peak versus amplitude. It is essential to separate three properties of the spectrum, only one of which is M -independent. (i) *Shape.* The dimensionless spectral form — the causal infrared slope $\Omega_{\text{GW}} \propto p^3$ (Sec. I C) and the inertial power law — is universal: it is set by causality and by the source power law, not by M . This is the property that is legitimately called “ M -independent,” and it is what the M^{-3} -compensated collapse of Fig. 1(b) isolates. (ii) *Peak location.* Here the literature genuinely splits. In spatial-scale-anchored treatments the peak is pinned to the source scale k_0 (set by the largest eddy / bubble size), so $p_{\text{peak}} = \mathcal{O}(1)$ and M controls only the amplitude [4]; in sweeping-anchored treatments the peak is tied to the eddy-turnover frequency $u_0 k_0 \propto M k_0$, so it shifts with M [2, 6]. Our kernel realises the second picture through $\eta_k \propto M k^{2/3}$, giving $p_{\text{peak}} \sim M$ (Sec. II C). The two are not in contradiction: a peak “at the stirring scale” is M -independent only because it is measured in units of k_0 and the source is assumed quasi-static over an eddy. (iii) *Total energy / amplitude.* This is never M -independent. In the stationary kernel the infrared amplitude scales as the M^3 prefactor of Eq. (21) ($\Omega_{\text{GW}} \rightarrow C M^3 p^3$, Sec. II C); more generally the quadrupole argument gives $\Omega_{\text{GW}} \propto \Omega_K^2 \propto M^4$, steepening toward M^{5-6} once the finite source duration is included [4], the extra power being absent from a single-time stationary kernel. Consequently a stated “ M -independence” of the GW signal can only refer to the shape (i); any claim that the peak position and the total energy are simultaneously M -independent is either a normalised-units artifact (dividing by Ω_K^2 and measuring p in units of k_0) or a conflation of the universal shape with the dimensional peak and amplitude.

III. MONOCHROMATIC AND WHITE-NOISE SPECTRA

We now consider two alternative models for the turbulence power spectrum and carry out the same GW-kernel derivation in each case. The first model replaces the Kolmogorov spectrum by a monochromatic (delta-function) distribution in wavenumber; the second takes the real-space two-point function to be spatially uncorrelated (delta function in $\mathbf{r}$), which corresponds to a flat (white-noise) spectrum in k -space. For each spatial model we work out both the Kraichnan stationary decorrelation and the decaying-turbulence temporal model, giving four cases in total.

The derivation follows the same sequence of steps as in Sec. II: (i) write H_{ijij} as a double convolution in both $\mathbf{k}$ and ω , (ii) insert the spectral model, (iii) perform the tensor contractions to get $\mathcal{K}$, (iv) switch to spherical coordinates and the variable $u = |\mathbf{k} - \mathbf{k}_1|$, (v) evaluate or simplify the temporal integral, and (vi) introduce dimensionless variables.

A. Monochromatic energy spectrum $E(k) = E_0 \delta(k - k_0)$

Set $E(k) = E_0 \delta(k - k_0)$ so that $A(k) = E_0 \delta(k - k_0)/(4\pi k^2)$ and the spectral tensor reads

$$\tilde{F}_{ij}(\mathbf{k}, \omega) = \frac{E_0 \delta(k - k_0)}{4\pi k^2} P_{ij}(\hat{k}) \tilde{f}(\omega), \quad \int_0^\infty E(k) dk = E_0, \quad (24)$$

with $\tilde{f}(\omega)$ to be specified by the decorrelation model.

Starting from Eq. (13) in spherical coordinates, the radial integral collapses $k_1 \rightarrow k_0$ via $\delta(k_1 - k_0)$, fixing $u = \sqrt{k^2 + k_0^2 - 2kk_0\mu}$. Trading μ for u (azimuth contributes 2π) and using $\delta(u - k_0)$ from $A(u)$ yields the closed-form

spatial reduction

$$2\pi \int_{-1}^1 d\mu A(u) = \frac{E_0}{2kk_0^2} \Theta(2k_0 - k), \quad (25)$$

with the Heaviside factor enforcing the triangle inequality $p \equiv k/k_0 \leq 2$. The geometric kernel at $k_1 = u = k_0$ evaluates to

$$\mathcal{K}_0(p) \equiv \mathcal{K}(k, k_0, k_0) = \frac{14}{3} - \frac{p^2}{3} + \frac{p^4}{12}. \quad (26)$$

Substituting Eqs. (25)–(26) in Eq. (13),

$$H_{ijij}(\mathbf{k}, \omega) = \frac{E_0^2 \Theta(2k_0 - k)}{2(2\pi)^8 k k_0^2} \mathcal{K}_0(p) I_\omega(\omega), \quad I_\omega(\omega) \equiv \int d\omega_1 \tilde{f}(\omega_1) \tilde{f}(\omega - \omega_1). \quad (27)$$

All spatial structure has been resolved in closed form; only the temporal convolution I_ω remains. Because both wavevectors are pinned to k_0 by the two δ -functions, there is no leftover wavenumber integration: the only spatial constraint is the M -independent triangle inequality $\Theta(2 - p)$. The Mach number therefore cannot reshape any spatial integration *bounds* here; it enters solely through the temporal decorrelation rate at the fixed scale k_0 , which we tie to M via $\eta_{k_0} = \frac{M}{\sqrt{2\pi}} k_0$ (cf. Eq. (19)). This contrasts with the white-noise case of Sec. III B, where M does enter the surviving wavenumber integral.

1. Kraichnan decorrelation

For $\tilde{f}(\omega) = (2/\eta_0) e^{-\omega^2/\pi\eta_0^2}$ with $\eta_0 \equiv \eta_{k_0} = Mk_0/\sqrt{2\pi}$, the Gaussian convolution gives $I_\omega(\omega) = (2\sqrt{2\pi}/\eta_0) e^{-\omega^2/(2\pi\eta_0^2)}$. With $q \equiv \omega/k_0$ the exponent becomes $\omega^2/(2\pi\eta_0^2) = q^2/M^2$, so Eq. (27) separates into a dimensional prefactor and the boxed dimensionless kernel

$$H_{ijij}(\mathbf{k}, \omega) = \frac{\sqrt{2} E_0^2}{(2\pi)^7 k k_0^2 \eta_0} \mathcal{K}_0(p) \Theta(2 - p) e^{-q^2/M^2}, \quad \mathfrak{H}_{\text{Kraichnan}}^{(\delta k)}(p, q; M) = \frac{\mathcal{K}_0(p)}{p} \Theta(2 - p) e^{-q^2/M^2}. \quad (28)$$

There is no remaining integral. The p -shape is fixed; M acts only as the Gaussian width in q , so the GW spectrum peaks at $q \sim M$.

2. Decay model

For decaying turbulence, $\tilde{f}(\omega) = \hat{g}(\omega\tau_1)$ with $\hat{g}(\bar{q}) = e^{i\bar{q}}(-i\bar{q})^{-5/3}\Gamma(\frac{1}{3}, -i\bar{q})$ and $\tau_1 = 1/\eta_0 = \sqrt{2\pi}/(Mk_0)$ the decay time at scale k_0 . With $q = \omega/k_0$ the natural temporal argument is $\bar{q} \equiv \omega\tau_1 = \sqrt{2\pi} q/M$; changing variables in Eq. (27), the temporal factor becomes $I_\omega = (1/\tau_1) \int dq_1 \hat{g}(q_1) \hat{g}(\bar{q} - q_1)$, which has no closed form. The dimensionless result is

$$\mathfrak{H}_{\text{decay}}^{(\delta k)}(p, q; M) = \frac{\mathcal{K}_0(p)}{p} \Theta(2 - p) \int_{-\infty}^{\infty} dq_1 \hat{g}(q_1) \hat{g}(\bar{q} - q_1), \quad \bar{q} = \frac{\sqrt{2\pi} q}{M}. \quad (29)$$

The spatial part is fully resolved by the two δ -functions; only a 1-D frequency integral remains, and M enters it solely through the rescaled frequency $\bar{q}$.

B. Real-space delta-function correlations

Take the spatial part of R_{ij} to be a $\delta^{(3)}(\mathbf{r})$, with velocity variance v_0^2 and temporal decorrelation $T(\eta_k, \tau)$:

$$R_{ij}(\mathbf{r}, \tau) = v_0^2 \delta^{(3)}(\mathbf{r}) P_{ij} T(\eta_k, \tau), \quad \tilde{F}_{ij}(\mathbf{k}, \omega) = v_0^2 P_{ij}(\hat{k}) \tilde{T}(\eta_k, \omega), \quad (30)$$

so $A(k) = v_0^2$ is k -independent (i.e. $E(k) \propto k^2$, ultraviolet-dominated); the integrals must be cut off between k_0 and k_d . Crucially, we adopt the same k -dependent sweeping rate as in Sec. II, $\eta_k = \frac{1}{\sqrt{2\pi}} \varepsilon^{1/3} k^{2/3} = \frac{M}{\sqrt{2\pi}} k_0^{1/3} k^{2/3}$,

so that $\tilde{T}$ depends on the wavenumber and the temporal factor no longer decouples from the spatial integral. With $A(k_1) = A(u) = v_0^2$ in Eq. (11) and dimensionless variables $z = k_1/k_0$, $y = u/k_0$, $p = k/k_0$, $q = \omega/k_0$, $R^{3/4} = k_d/k_0$,

$$H_{ijij}(\mathbf{k}, \omega) = \frac{v_0^4 k_0^2}{(2\pi)^7} \int_1^{R^{3/4}} \frac{z dz}{p} \int_{\max(|p-z|, 1)}^{\min(p+z, R^{3/4})} y dy \tilde{\mathcal{K}}(p, z, y) J(\omega; k_1, u), \quad J(\omega; k_1, u) \equiv \int d\omega_1 \tilde{T}(\eta_{k_1}, \omega_1) \tilde{T}(\eta_u, \omega - \omega_1), \quad (31)$$

where $\tilde{\mathcal{K}}(p, z, y) = \mathcal{K}(k, k_1, u)|_{k_1=z k_0, u=y k_0}$ and the frequency factor J now carries the k_1, u dependence inherited from η_{k_1}, η_u .

1. Kraichnan decorrelation

With $\tilde{T}(\eta_k, \omega) = (2/\eta_k) e^{-\omega^2/\pi\eta_k^2}$ the convolution combines two Gaussians of *different* widths η_{k_1}, η_u ; using $\int d\omega_1 e^{-a\omega_1^2} e^{-b(\omega-\omega_1)^2} = \sqrt{\pi/(a+b)} e^{-ab\omega^2/(a+b)}$ with $a = 1/\pi\eta_{k_1}^2$, $b = 1/\pi\eta_u^2$, and $\eta_{k_1}^2 + \eta_u^2 = \frac{M^2}{2\pi} k_0^{2/3} (k_1^{4/3} + u^{4/3})$, the frequency factor becomes

$$J(\omega; k_1, u) = \frac{4\pi\sqrt{2\pi}}{M k_0^{1/3} \sqrt{k_1^{4/3} + u^{4/3}}} \exp\left(-\frac{2\omega^2}{M^2 k_0^{2/3} (k_1^{4/3} + u^{4/3})}\right). \quad (32)$$

Inserting this into Eq. (31) and absorbing the constants into the dimensional prefactor gives the boxed dimensionless kernel

$$\mathfrak{H}_{\text{Kraichnan}}^{(\delta r)}(p, q; M, R) = \int_1^{R^{3/4}} \frac{z dz}{p} \int_{\max(|p-z|, 1)}^{\min(p+z, R^{3/4})} y dy \tilde{\mathcal{K}}(p, z, y) (z^{4/3} + y^{4/3})^{-1/2} e^{-2q^2/[M^2(z^{4/3} + y^{4/3})]}. \quad (33)$$

The exponential now acts as an M -dependent effective cutoff on the wavenumber integration: contributions require $z^{4/3} + y^{4/3} \gtrsim 2q^2/M^2$, so as M grows the effective wavenumber support widens and the GW spectrum peak shifts. This is precisely the structure of the Gogoberidze exponent $\exp[-2\epsilon^{-2/3}\omega^2/(k_1^{4/3} + u^{4/3})]$ in Eq. (18); the k -dependent sweeping rate, rather than a fixed η_0 , is what makes M control the integration support.

2. Decay model

For decaying turbulence $\tilde{T}(\eta_k, \omega) = \hat{g}(\omega\tau_k)$ with the same $\hat{g}(q) = e^{iq}(-iq)^{-5/3}\Gamma(\frac{1}{3}, -iq)$ as Sec. III A 2, but now with the k -dependent decay time $\tau_k = 1/\eta_k = \sqrt{2\pi}/(M k_0^{1/3} k^{2/3})$. The temporal factor is the convolution $J(\omega; k_1, u) = \int d\omega_1 \hat{g}(\omega_1\tau_{k_1}) \hat{g}((\omega - \omega_1)\tau_u)$, which has no closed form and, because τ_{k_1}, τ_u depend on the wavenumbers, must remain *inside* the spatial integral. Writing $\tilde{J}(q; z, y, M)$ for this convolution with the per-mode dimensionless frequencies fixed by $\tau_{k_1}, \tau_u \propto 1/M$, the boxed result is

$$\mathfrak{H}_{\text{decay}}^{(\delta r)}(p, q; M, R) = \int_1^{R^{3/4}} \frac{z dz}{p} \int_{\max(|p-z|, 1)}^{\min(p+z, R^{3/4})} y dy \tilde{\mathcal{K}}(p, z, y) \tilde{J}(q; z, y, M). \quad (34)$$

The decay rates scale as $\eta_k \propto M k_0^{1/3} k^{2/3}$, so M again sets the effective wavenumber support. Unlike the fixed- η_0 case, the three integrals no longer factorize: the k -dependence of the decay rate couples the frequency convolution to the wavenumber integration, mirroring the behavior of Eq. (18).

Summary of the four dimensionless integrals

Spectrum	Decorrelation	Dimensionless integral
$E_0\delta(k - k_0)$	Kraichnan	Eq. (28): closed form; M in frequency scale only
$E_0\delta(k - k_0)$	Decay	Eq. (29): 1D frequency integral; M in $\bar{q}$
$\delta^{(3)}(\mathbf{r})$	Kraichnan	Eq. (33): 2D wavevector integral; M -cutoff in bounds
$\delta^{(3)}(\mathbf{r})$	Decay	Eq. (34): 2D wavevector $\times$ nested 1D frequency; M -cutoff

IV. NUMERICAL RESULTS: $f_{\text{peak}}^{\text{GW}}(\tau_*, M_0)$

We scan the peak GW frequency $f_{\text{peak}}^{\text{GW}} = \text{argmax}_{\omega} [\omega \Omega_{\text{GW}}(\omega)]/2\pi$ over the (τ_*, M_0) plane for a delta-shell decaying source evolved with the BK2016 self-similar closure and time-integrated over the emission window (`Notebooks/f_peak_vs_tau_mach`). Two temporal-correlation kernels are compared through their dimensionless transform $\mathcal{G}(q) = \widehat{R}^2(q)$: the random-phase BK2016 form (Model A) and the coherent eddy (Model B), shown in Fig. 3.

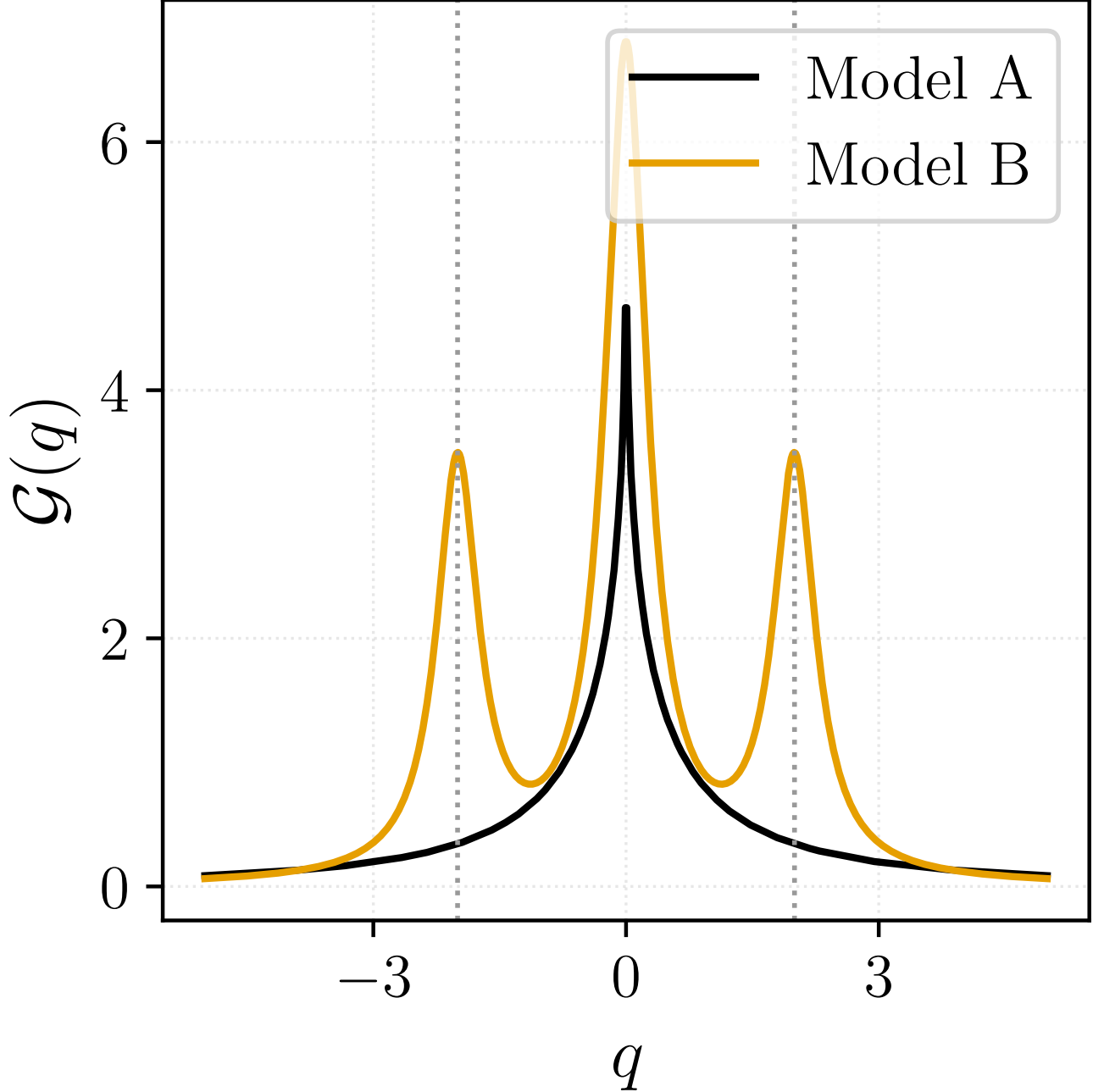


FIG. 3. Dimensionless temporal kernel $\mathcal{G}(q) = \widehat{R}^2(q)$ for the two correlation models considered in Sec. IV. Model A uses the random-phase BK2016 form $R_A(\sigma) = (1 + |\sigma|)^{-2/3}$; Model B is the coherent eddy $R_B(\sigma) = \cos(\sigma) e^{-\alpha|\sigma|}$ with $\alpha = 0.15$.

Across all four self-similar classes the ratio $f_{\text{peak}}^A/f_{\text{peak}}^B$ is *independent of* (τ_*, M_0) : the self-similar closure makes both the eddy frequency ω_e and the sweeping rate η_0 scale as $(1 + T/\tau_*)^{-1}$, so the two kernels inherit the same overall

frequency scaling and differ only in their fixed dimensionless peak position. The model difference therefore collapses to a single per-class constant (Fig. 4): $f_A/f_B \simeq 0.37, 0.36$ for the HD (Loitsiansky, Saffman) classes and $0.83, 0.78$ for the MHD (nonhelical, helical) classes – Model A always peaks lower. The large HD offset comes from the side-peak of the coherent kernel at the eddy frequency ($q \simeq \pm 2$, the $\cos \sigma$ oscillation of R_B), which wins the $\omega \Omega_{\text{GW}}$ argmax for the steep-decay HD classes but nearly coincides with the broad Model-A peak for MHD. Because the per-model maps differ only by this constant, we display the ratio rather than the eight individual heatmaps.

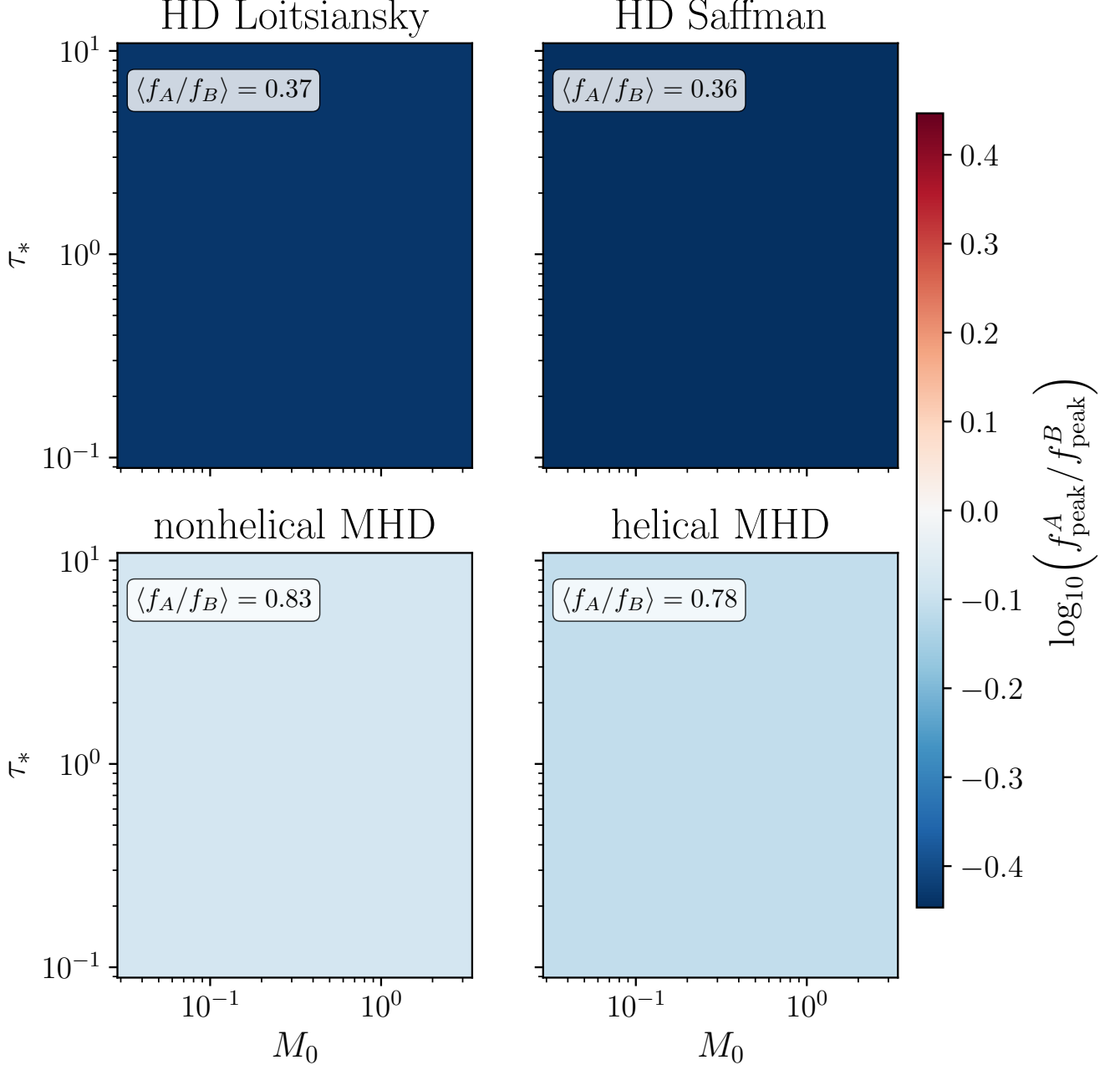


FIG. 4. Model A/B ratio of the peak GW frequency, $\log_{10}(f_{\text{peak}}^A / f_{\text{peak}}^B)$, on the (τ_*, M_0) plane for the four BK2016 self-similar decay classes (HD Loitsiansky $q = 2/7$, $\beta = 4$; HD Saffman $q = 1/3$, $\beta = 3$; nonhelical MHD $q = 1/2$, $\beta = 1$; helical MHD $q = 2/3$, $\beta = 0$). Each panel is flat: the ratio is independent of (τ_*, M_0) , fixed only by the kernel shapes, with the per-class value $\langle f_A/f_B \rangle$ annotated. Model A (random-phase) peaks below Model B (coherent eddy) everywhere; the HD classes show the larger offset.

V. SPECTRUM SURVEY FOR ALL DERIVED MODELS

In this section we plot, on a common convention, the dimensionless GW energy-density spectrum

$$\Omega_{\text{GW}}(p) \propto p^3 H(p, p), \quad p = k_{\text{GW}}/k_0, \quad (35)$$

along the sound-cone diagonal $q = p$ for every model derived in this paper. The p^2 factor relative to the characteristic strain $h_c^2(p) = p H(p, p)$ is the phase-space measure d^3k ; together with causality and a finite white-noise source it predicts $\Omega_{\text{GW}} \sim p^3$ in the IR, followed by a peak near $p \sim 1$ and a decay in the UV. A dotted grey line shows the p^3 reference; cases whose IR slope deviates from 3 by more than ± 0.5 are flagged in the build log of `Notebooks/all_spectra.py`. Models with a $1/p$ pole at the origin (monochromatic $E_0 \delta(k - k_0)$ source) cannot satisfy the universal scaling because the fluid itself carries no IR power.

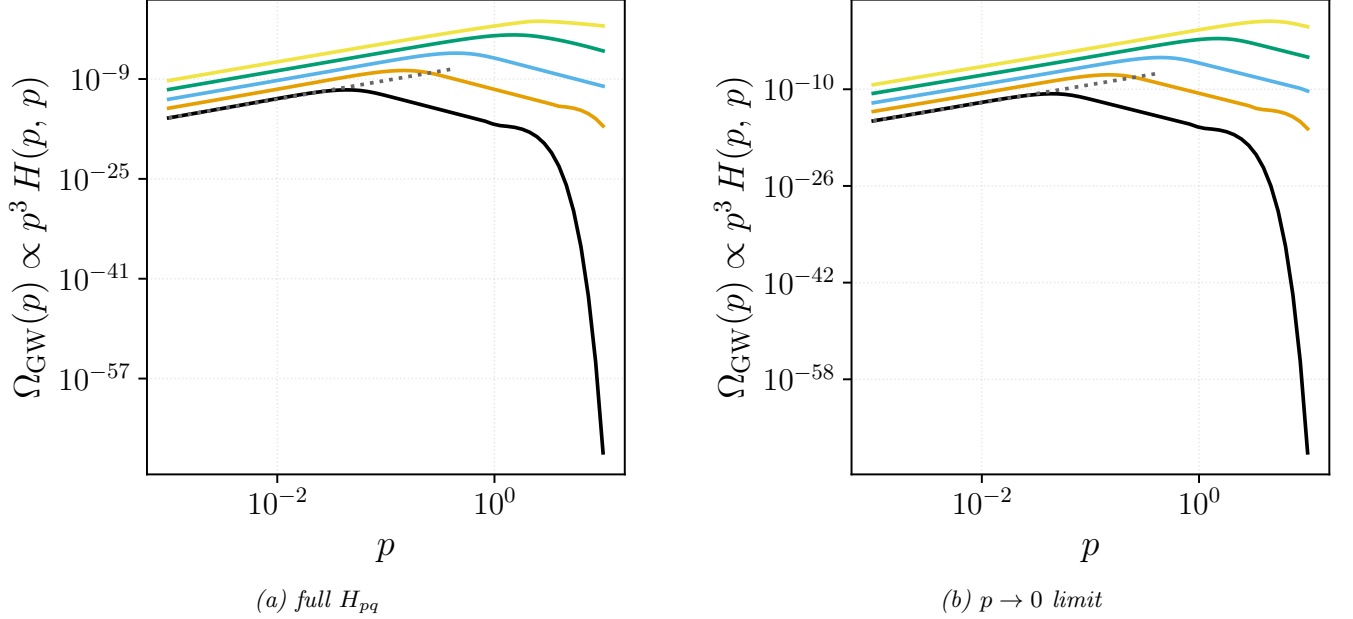


FIG. 5. Kraichnan decorrelation + Kolmogorov spectrum. Five curves per panel correspond to $M = 0.03, 0.1, 0.3, 1.0, 3.0$ at $R = 10^4$. Panel (a) is the full numerical H_{pq} (Sec. II); panel (b) is the aeroacoustic $p \rightarrow 0$ limit (Kahniashvili et al. 2008). Both give the universal IR slope $\Omega_{\text{GW}} \propto p^3$.

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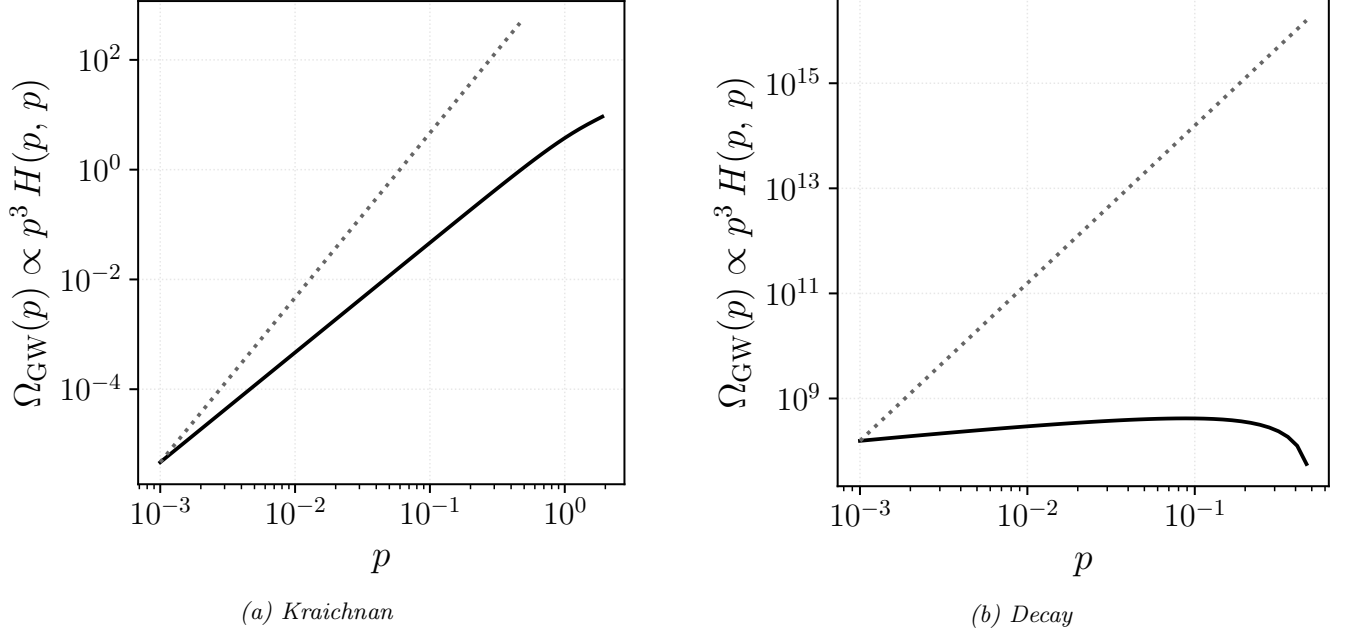


FIG. 6. Monochromatic spectrum $E(k) = E_0 \delta(k - k_0)$ (no free parameters at the dimensionless level). Panel (a): Kraichnan decorrelation, closed form $\Omega_{\text{GW}}(p) \propto p^3 K_0(p)/p \Theta(2-p) e^{-p^2/(2\pi)}$ with IR slope **2** (flagged: $1/p$ pole from the monochromatic source removes one power of p from the universal scaling). Panel (b): decay temporal convolution, IR slope **0.3** (flagged: the temporal-tail extra suppression further reduces the IR power).

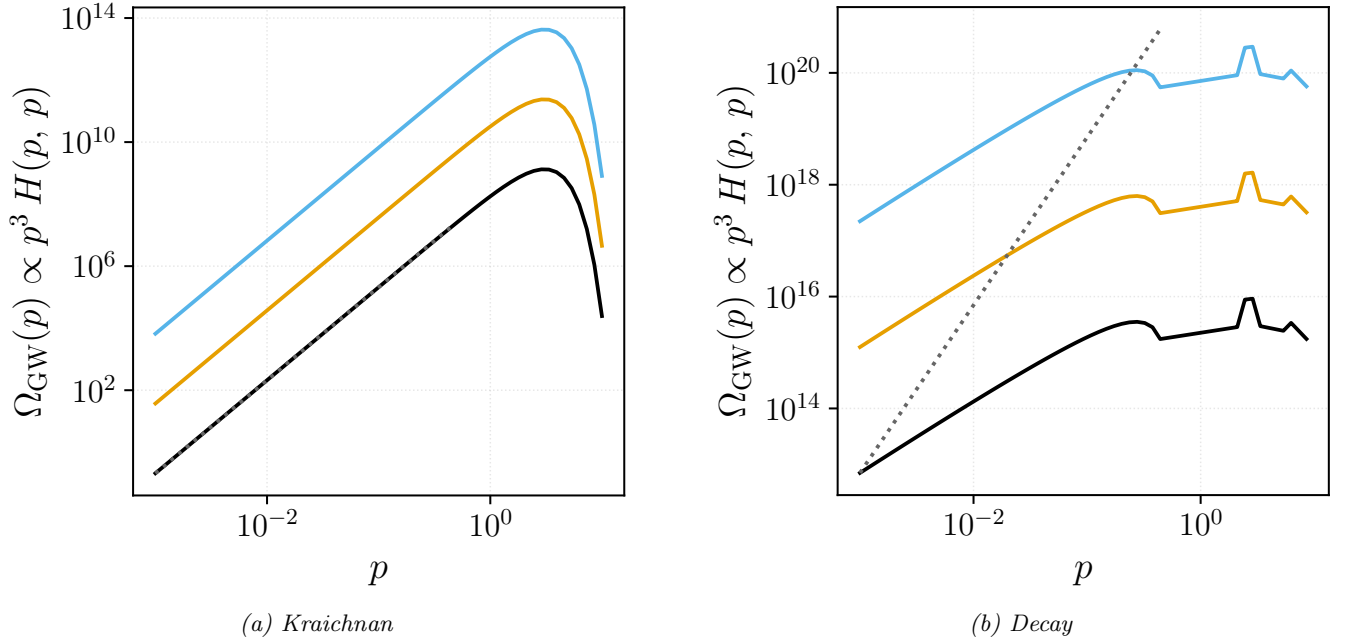


FIG. 7. White-noise real-space correlations $R_{ij} \propto \delta^{(3)}(\mathbf{r})$. Three curves per panel correspond to $R = 10^3, 10^4, 10^5$. Panel (a): Kraichnan decorrelation, matches the universal IR slope **3**. Panel (b): decay temporal model, IR slope **1.3** (flagged: temporal-convolution tails pull power down at small p).